American Terawatt

TEMPLATE TEST

Cable Link Single-Line Diagram

TikZ rendering of the sending-end / receiving-end transmission model

Test document for the shared report template. It compiles a TikZ rendering of the single-line diagram of a cable transmission link (source, disconnectors, cable of length L_{cable} and load impedance Z_L) with the sending-end and receiving-end voltages, currents and power flows.

Revision history

REV.	DATE	AUTHOR	DESCRIPTION
0.1	Sep. 18, 2026	Eugen Starschich	Initial document

1 Transmission model

Figure 1 shows the single-line representation of a cable transmission link between a sending end S and a receiving end R. The sending-end source feeds the cable through a disconnector; at the receiving end a second disconnector connects the cable to the load impedance Z_L. The voltages U_S and U_R are measured against ground at the two cable ends, the currents I_S and I_R are counted positive in the direction from S to R, and P_S, Q_S and P_R, Q_R denote the active and reactive power entering and leaving the cable of length L_{cable}.

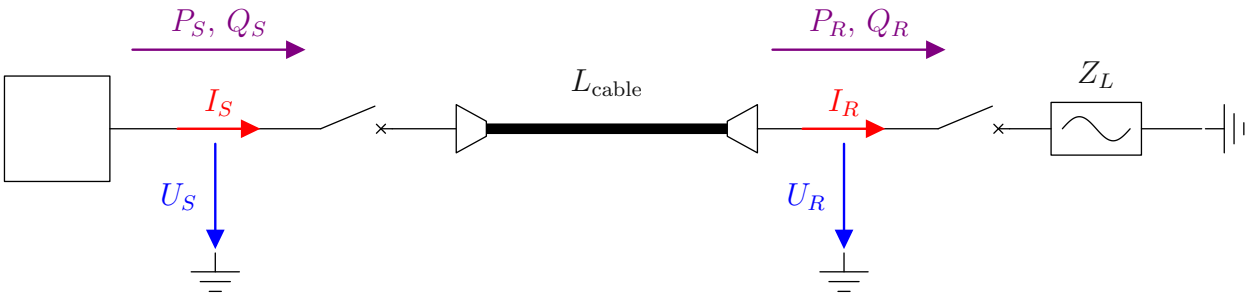


Figure 1: Single-line diagram of the cable transmission link (TikZ).

The figure is drawn in TikZ (`figures/transmission-line-tikz.tex`) and can be reused in other documents with `\input`.

2 Comparison with the original

Figure 2 reproduces the original drawing (`figures/image9.pdf`) that the TikZ figure was traced from, for a side-by-side check of geometry, colours and labels. Both figures are scaled to the full text width, so they can be compared directly.

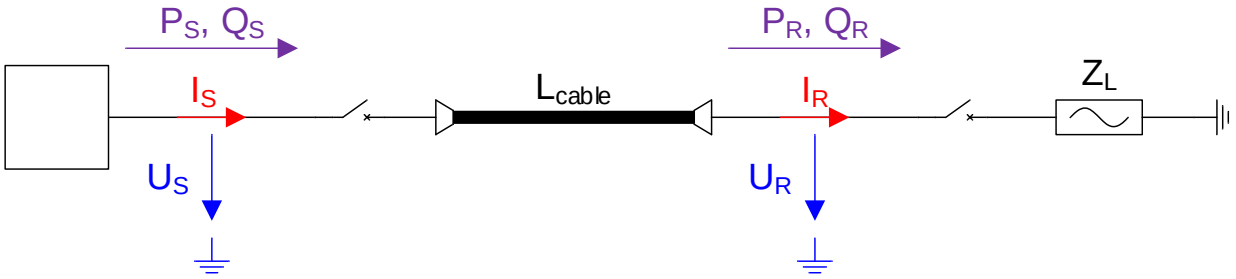


Figure 2: Original drawing used as reference.

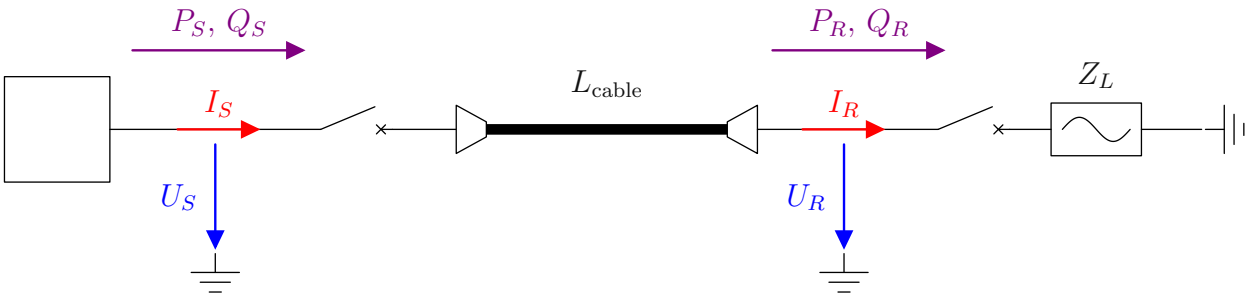


Figure 3: TikZ rendering of Figure 2.